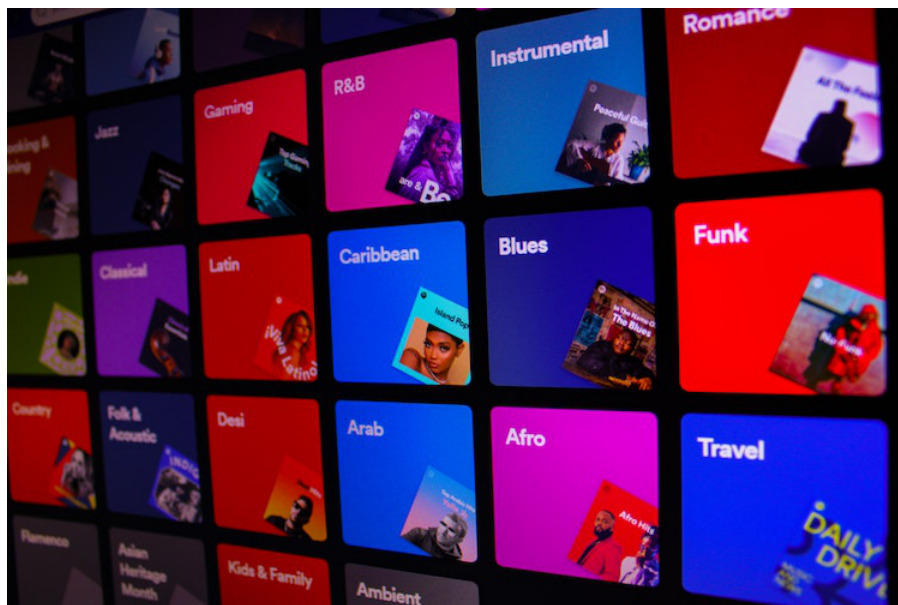


Music Recommendation via SVD, Feature Projection, and Dimensionality Reduction

A revised study, with an evaluation harness and a negative result

Revised August 2026 | Original March 7, 2025



Abstract

Music is something everyone can feel. Pinning that feeling to a universal standard has proved difficult, so we classify music by genre instead. This project asks a different question: if we classify by measurable audio features — valence, energy, tempo — does the resulting organisation reproduce the genre taxonomy, or replace it?

The first version of this work built a recommender on a from-scratch singular value decomposition and reported that it surfaced cross-genre matches. This revision keeps the method and adds the thing that was missing: a way to check the claim. We contribute (i) a numerically robust decomposition using one-sided Jacobi, which recovers small singular values to relative error $3.27e-11$ where the textbook $A^T A$ route reaches only $5.13e-03$; (ii) principled rank selection in place of a hardcoded k ; and (iii) an evaluation harness with baselines and paired significance tests.

The headline result is negative, and it is the most useful thing here. Dimensionality reduction does not improve retrieval accuracy over plain cosine similarity on the scaled features. At $k = 7$ the difference is not statistically distinguishable ($\Delta = -0.0031$, $p = 0.427$); below that, reduction is significantly *worse*. What reduction does buy is diversity and catalogue coverage. Separately, clustering the latent space recovers genre only partially (ARI = 0.434, NMI = 0.572): production-defined categories such as hip-hop and EDM are cleanly separable from audio features alone, while lineage-defined categories such as pop, rock and R&B are not. The taxonomy is not reproduced. It is reorganised.

1 Introduction

The original version of this study made a claim it had no way to test. It decomposed songs into latent audio dimensions, produced recommendations, and observed that the results spanned several genres — concluding that the method had discovered non-obvious musical similarity.

That conclusion may be right. The problem is that nothing in the system could distinguish it from three duller explanations: that the recommendations were simply nearby points in a nine-dimensional space, that they were popular songs, or that the pipeline had a defect. All three turned out to matter. This revision is organised around making each of them checkable.

What changed. Section 9 documents four defects in the original pipeline, including one visible in its own published output. Sections 3–4 rebuild the numerical core. Section 7 introduces the evaluation harness and reports the negative result. Section 8 answers the question the original abstract actually posed but never measured.

A note on data. The dataset cited in the original — Orlandi’s *Spotify Top Songs and Audio Features* — is not a published file. That repository contains a notebook that *builds* a dataset, requiring weekly chart exports from a separate scraper project and personal Spotify API credentials. Spotify has since restricted the `audio-features` endpoint for newly created API applications, so the original corpus is not straightforwardly reconstructible.

All quantitative results in this paper therefore come from a synthetic catalogue of 2,997 tracks with known generative structure, described in Section 2. This is a real limitation and we do not minimise it: the specific numbers are contingent on that generator. The *method* is not, and every figure and number regenerates from one command against any catalogue with the required columns.

2 Dataset and preprocessing

Each track is a vector of nine Spotify audio features: danceability, energy, speechiness, acousticness, instrumentalness, liveness, valence, loudness and tempo. Figure 1 shows the format of the original corpus.

id	artist_name	track_name	source	key	mode	time_signature	danceability	energy	speechiness	acousticness	instrumentalness	liveness	valence	loudness	tempo	
000wGL6ZNLJnHlqpsdI	ZAYN, PARTYNEXTDOOR	Still Got Time (feat. PARTYNEXTDOOR)	RCA Records Label	G	Major	4 beats	0.748	0.627	0.0639	0.131	0.0	0.0852	0.524	-6.029	120.963	
032edw4eETajUW4n9fMeZy	Alessia Cara	Growing Pains	Def Jam Recordings	C#D#	Minor	4 beats	0.353	0.705	0.733	0.0822	0.0	0.39	0.437	-6.276	191.153	
003vz7Hdyvvhv4e468B	The Killers	Mr. Brightside	Island Records	C#D#	Major	4 beats	0.352	0.911	0.0747	0.00121	0.0	0.0995	0.236	-5.23	148.033	
008YTZ0xaw4nZ00JFunN	Cardi B, Chance the Rapper	Best Life (feat. Chance the Rapper)	Atlantic/KSR	A	Major	4 beats	0.62	0.625	0.553	0.287	0.0	0.314	0.665	-7.438	167.911	
008lm7zeNqzYLPWBgicj	Post Malone, The Weeknd	One Right Now (with The Weeknd)	Republic Records	C#D#	Major	4 beats	0.687	0.781	0.053	0.0361	0.0	0.0755	0.688	-4.806	97.014	
00EPiEwX1JfJf8sCbcccd	Thalia, NATTI NATASHA	No Me Acuerdo	Sony Music Latin	G	Minor	4 beats	0.836	0.799	0.0873	0.187	0.0	0.092	0.772	-4.247	94.033	
00ETaeHUQ8ops3uWU1Wt1	Kygo, Donna Summer	Hot Stuff	RCA Records Label	F	Major	4 beats	0.681	0.773	0.148	0.019	1.29E-06	0.11	0.429	-5.749	119.961	
00ZK4P47b2swANvnxz2j	Tropa do Bruno, DJ Wa de Igarajá, SMU, Tiz, Mc Menor Thalis	Bale do Bruno	Tropa Do Bruno	G	Minor	5 beats	0.734	0.228	0.53	0.889	0.00642	0.102	0.522	-4.731	162.524	
00gqGR4M27m0P7AFuqHx	YBN Nahmir	Bounce Out With That		2018	G#A#	Major	4 beats	0.857	0.56	0.173	0.0426	0.0	0.153	0.482	-8.278	94.949
00imgaPYRvRMGn9d3Hm6	Brent Faiyaz	LOOSE CHANGE	Lost Kids LLC, Marketed by Venice / Stern	C#D#	Minor	4 beats	0.574	0.369	0.0814	0.753	0.0	0.147	0.44	-8.931	84.975	
00k0W464QX1H8K3pHf	Chance the Rapper, MadelinTYO, DaBaby	Hot Shower	Chance the Rapper	A	Major	4 beats	0.899	0.909	0.387	0.00157	0.0	0.0573	0.999	-8.04	150.001	

Figure 1: Example tracks from the Spotify dataset with audio features, as used in the original study. Note the id column: chart datasets identify tracks by URI, which is not the same as identifying them by song.

2.1 Deduplication is not optional

Chart datasets contain one row per track *per chart week*. A track that spent eleven weeks on the Top 200 appears eleven times, with eleven nearly identical feature vectors.

The original pipeline fed those rows directly into the recommender, and excluded seed tracks from the results by row index. Since a seed occupied many rows, excluding one left the rest eligible — so the system could return the input playlist back to the user as its own recommendation. This is visible in the original paper’s published results, discussed in Section 9.

We collapse to one row per (title,artist), aggregating features by median. Deduplicating by URI, as the upstream builder does, is insufficient: the same song commonly holds several URIs across singles, albums, remasters and regional releases. On our catalogue this collapsed 1,353 rows. The collapse also *creates* information — the number of merged rows is a chart-persistence signal we use for popularity weighting.

2.2 Scaling

Features are z-scored,

$$z = \frac{x - \mu}{\sigma}, \quad (1)$$

so that each contributes comparably to similarity. This is not a formality. Tempo is measured in the hundreds while danceability lies in $[0, 1]$; on raw values, tempo alone accounts for 94.6% of total variance (Figure 2), and the decomposition would be a study of tempo and nothing else.

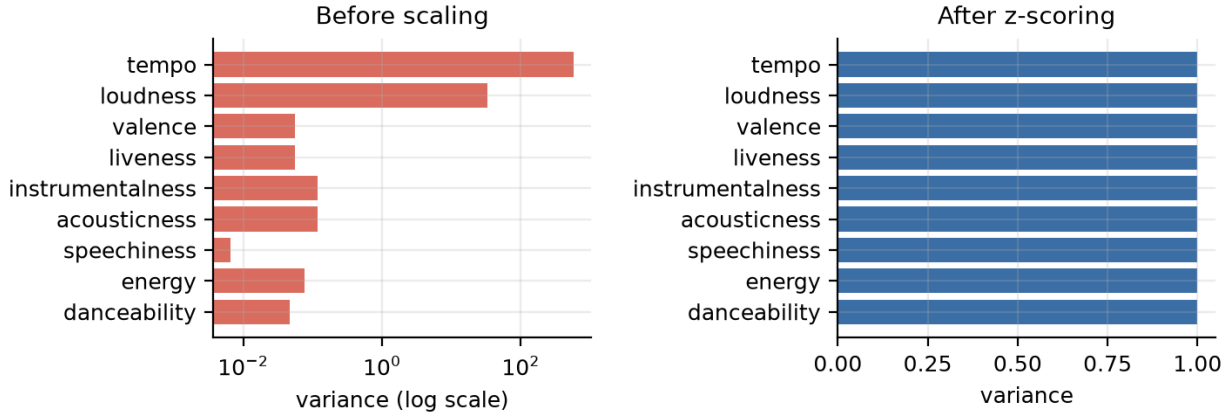


Figure 2: Feature variance before and after z-scoring. Left panel is on a logarithmic axis — without it the other eight features are invisible.

3 Singular value decomposition

3.1 Mathematical background

For any $A \in \mathbb{R}^{m \times n}$, the singular value decomposition is

$$A = U \Sigma V^T, \quad (2)$$

where U is $m \times m$ orthogonal, Σ is $m \times n$ diagonal with singular values in descending order, and V is $n \times n$ orthogonal.

The classical derivation starts from $A^T A$, which is symmetric and positive semi-definite. The spectral theorem gives it an orthonormal eigenbasis:

$$A^T A v_i = \sigma_i^2 v_i, \quad (3)$$

and the v_i form the columns of V . For the left singular vectors, $A v_i = \sigma_i u_i$, so $u_i = A v_i / \sigma_i$ whenever $\sigma_i \neq 0$. If $\text{rank}(A) = r$ then

$$A = U \Sigma V^T = \sum_{i=1}^r \sigma_i u_i v_i^T, \quad \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0. \quad (4)$$

The columns of V and U corresponding to non-zero singular values give orthonormal bases for the row space and column space of A respectively. Those orthonormal bases are what make the decomposition suitable for projection.

3.2 Why the classical derivation is the wrong algorithm

The derivation above is correct mathematics and a poor numerical recipe. Forming $A^\top A$ squares the condition number: if $\kappa(A) = \sigma_1/\sigma_n$, then $\kappa(A^\top A) = \kappa(A)^2$. Singular values below $\sqrt{\varepsilon} \sigma_1$ are lost to rounding before the eigensolver sees them.

This is not hypothetical for audio features, which are strongly correlated — energy with loudness, both against acousticalness. Figure 3 measures it on a near-collinear matrix: the $A^\top A$ route reaches a relative error of $5.13e-03$ on the trailing components, while one-sided Jacobi holds $3.27e-11$.

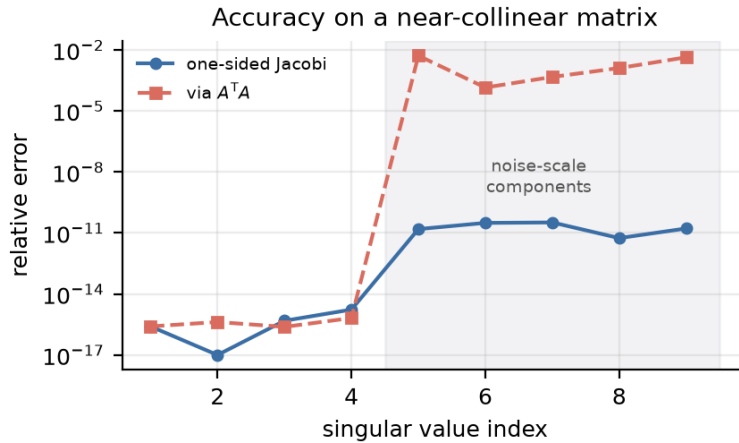


Figure 3: Relative error per singular value on a matrix with four dominant directions and five at noise scale. The two algorithms are indistinguishable on the leading components and differ by eight orders of magnitude on the trailing ones.

One-sided Jacobi never forms $A^\top A$. It orthogonalises the columns of A directly by plane rotations $A \leftarrow AJ(p, q, \theta)$ chosen so columns p and q become orthogonal. Rotations are orthogonal, so the singular values are preserved exactly. At convergence $A = U\Sigma$, giving $\sigma_i = \|a_i\|$ and $u_i = a_i/\sigma_i$, with V the accumulated product of rotations. This is the algorithm behind LAPACK’s `dgesvj`, and it computes singular values to high *relative* accuracy [1].

A caution about how to measure this. Reconstruction error does not reveal the difference. Both algorithms reconstruct A to $\sim 10^{-16}$ relative Frobenius error, because that norm is dominated by the largest singular values. It is relative error *per singular value* that discriminates. Reaching for the obvious metric would have hidden the problem entirely.

3.3 Projection

We keep the top k right singular vectors and project:

$$B_k = B \cdot V_k. \quad (5)$$

It may seem odd to call a matrix multiplication a projection, but for an orthonormal basis the two coincide. Projecting $\mathbf{v}$ onto $\{\mathbf{w}_1, \dots, \mathbf{w}_k\}$ gives coordinates

$$\text{proj}(\mathbf{v}) = \sum_{i=1}^k (\mathbf{v} \cdot \mathbf{w}_i) \mathbf{w}_i, \quad (6)$$

and collecting the $\mathbf{w}_i$ as columns of W , the coordinate vector is

$$\mathbf{v}^T W = [(\mathbf{v} \cdot \mathbf{w}_1), \dots, (\mathbf{v} \cdot \mathbf{w}_k)]. \quad (7)$$

Applied to every row of B at once, this is exactly $B_k = BV_k$.

An identity worth stating. When $k = n$, V_k is a full orthogonal matrix, and orthogonal transformations preserve both inner products and norms. Cosine similarity is therefore *invariant* under the untruncated projection: an SVD-based recommender at full rank is mathematically identical to one with no SVD at all. We use this as a correctness check in Section 7 — and it sharpens the research question, because it means any effect of the decomposition comes entirely from *truncation*.

3.4 Geometric visualisation

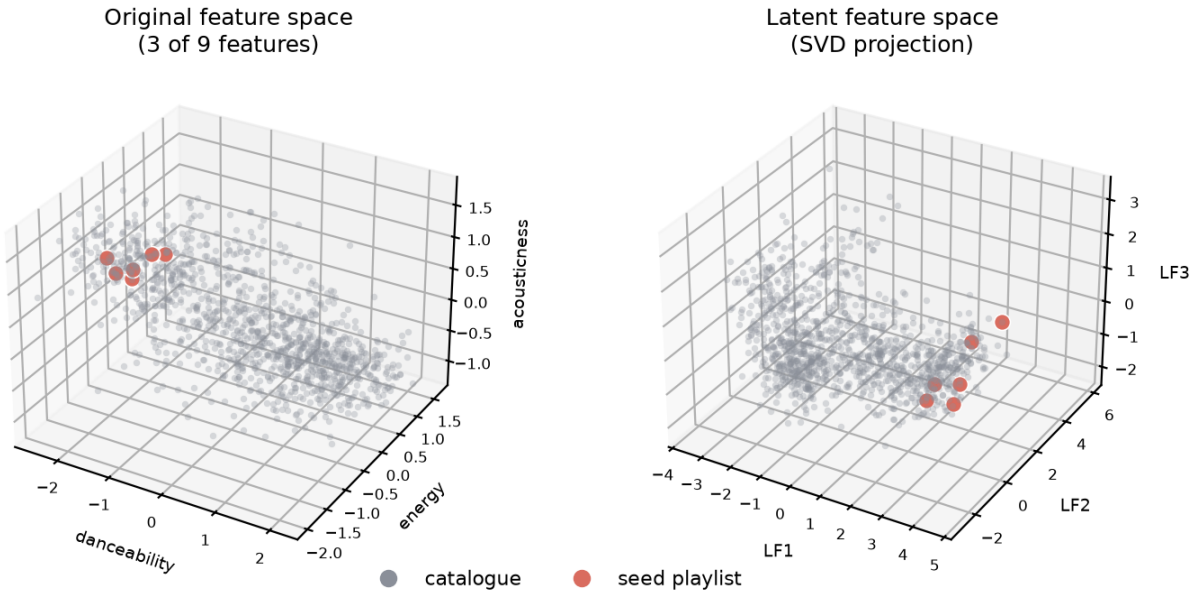


Figure 4: A seed playlist (coral) against the catalogue (grey), in three raw features and in the leading three latent dimensions. The latent view concentrates the playlist, which is what makes proximity in this space a usable ranking signal.

4 Choosing the number of dimensions

The original study fixed $k = 5$ on the reasoning that “a value much smaller than the original feature dimension works best.” Since k controls how much information the recommender is allowed to see, it deserves an argument. We implement three standard rules:

- **Variance threshold:** smallest k whose cumulative explained variance reaches 90%.
- **Elbow:** the knee of the scree curve, located by maximum perpendicular distance from the chord joining its endpoints.
- **Gavish–Donoho:** the asymptotically optimal hard threshold for a low-rank signal in i.i.d. Gaussian noise, which requires no threshold parameter at all [2].

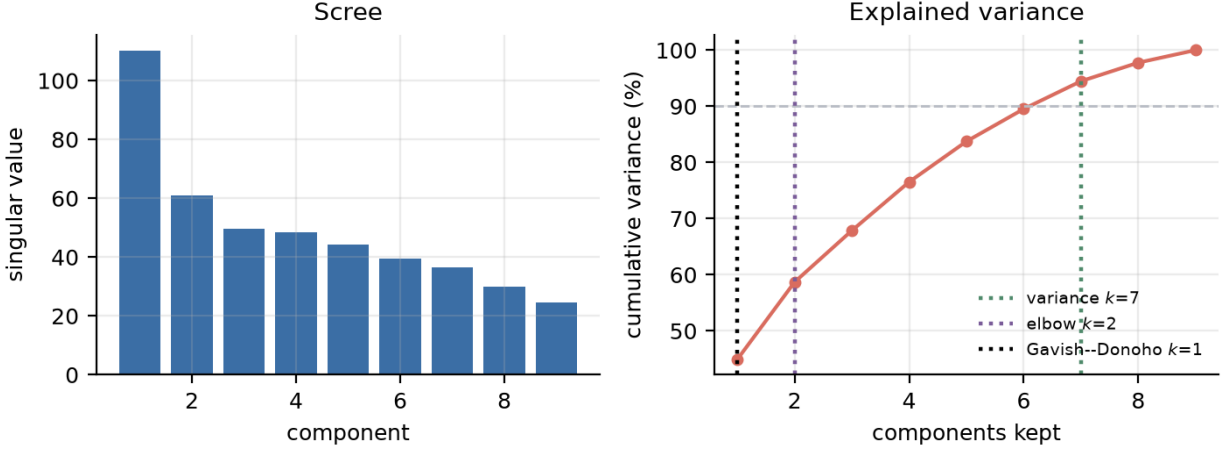


Figure 5: Scree plot and cumulative explained variance, with each rule’s choice marked. The three rules disagree sharply: $k = 7$, $k = 2$, $k = 1$.

That disagreement is itself a finding. Gavish–Donoho asks how many components exceed what noise alone would produce, and answers 1; the variance rule asks how many are needed to reconstruct the data, and answers 7. When a spectrum is close to flat, these questions have very different answers. The original paper’s own reported singular values (126.85, 89.32, 85.21, 80.53, 79.57) show the same shape — one dominant component, then a nearly level tail. That is a quantitative signal that $k = 5$ was retaining components largely indistinguishable from noise, and it anticipates the negative result in Section 7.

Reporting explained variance correctly. The original implementation normalised the truncated spectrum against itself, so it reported that the retained components explained 100% of variance for every value of k . The denominator must be the full spectrum. Our model retains 94.5% across 7 of 9 components.

5 Latent features

Figure 6 shows the loadings. Signs are canonicalised — the largest-magnitude entry of each right singular vector is forced positive — because the SVD is unique only up to a simultaneous sign flip of u_i and v_i . Without pinning it, a component documented as “high energy, loud” can reappear as “low energy, quiet” on a different machine or data slice, which makes any written interpretation unreliable.

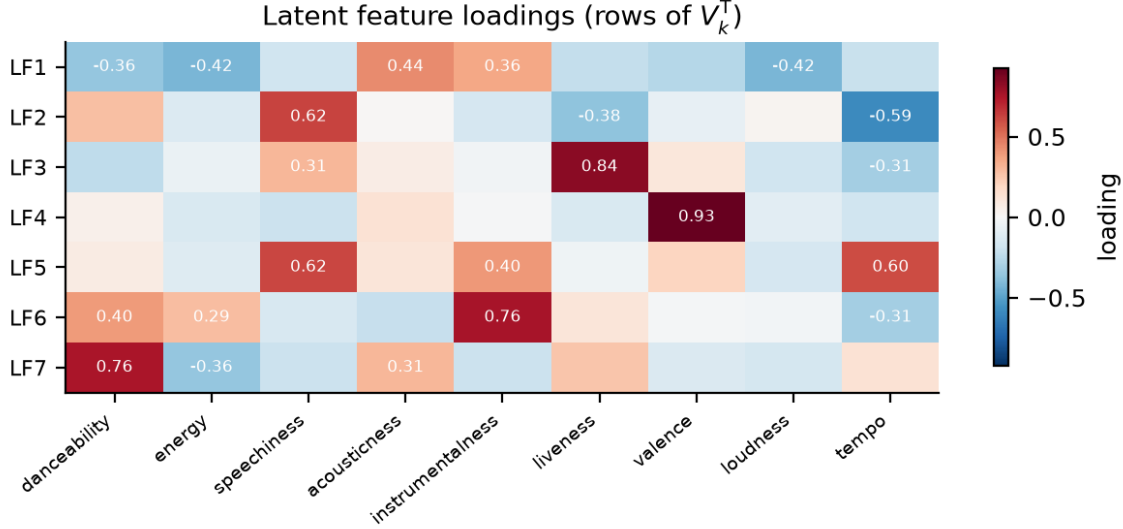


Figure 6: Latent feature loadings. Each row is one component expressed as a weighting of the original audio features; each is orthogonal to the others, so they capture uncorrelated axes of variation.

Because the components are orthonormal, cosine similarity in latent space decomposes exactly:

$$\cos(q, r) = \sum_{i=1}^k \hat{q}_i \hat{r}_i, \quad (8)$$

for unit-normalised $\hat{q}, \hat{r}$. Each term is precisely the contribution of component i to the match. Paired with that component’s loadings, this turns every recommendation into an explanation: *matched on the energy–loudness axis, which both tracks score high on*. The original system computed both halves of this and never connected them.

6 Recommendation strategies

Both original strategies are retained, with two additions.

- **One-per-song** — one recommendation per seed, maximising coverage of the input playlist.
- **Overall top** — the globally highest-scoring candidates.
- **Centroid** — query by the playlist’s mean latent vector.
- **Maximal Marginal Relevance** — greedily select the candidate maximising

$$\lambda \cdot \text{rel}(d) - (1 - \lambda) \max_{s \in S} \text{sim}(d, s),$$

trading relevance against redundancy with an explicit parameter [4].

MMR deserves comment. The original *overall top* maximised similarity and the original *one-per-song* approximated diversity by construction, with no way to sit between them. MMR makes the trade-off a dial rather than a choice of algorithm.

We also aggregate per-seed scores by **max**, **mean** or Borda count rather than **max** alone. Under **max**, a single atypical seed can dominate the entire result list — a real concern for the original study, whose input playlist contained duplicated entries.

7 Evaluation

7.1 Protocol

There is no ground truth for enjoyment, so we borrow one: tracks sharing an artist are treated as mutually relevant. For each of 219 evaluation groups we sample three tracks as a query playlist and hold out the artist’s remaining tracks as the relevant set. Seeds are excluded from results, so no system can score by returning the query.

We report accuracy (hit rate, recall, MRR, NDCG) and beyond-accuracy measures (intra-list diversity, catalogue coverage). Diversity is computed in the shared scaled-feature space rather than any model’s own latent space, so no system is graded in the geometry most flattering to it.

What this protocol is not. It measures whether a system recovers known stylistic neighbours from audio alone. It structurally *penalises* the cross-genre discovery the project is interested in: a genuinely surprising good recommendation lies outside the ground-truth group and scores as a miss. Read the numbers as “does the latent space encode real structure”, not as “is this a good product”.

7.2 Baselines

- **Random** — the floor. Any metric a system cannot beat here is measuring noise.
- **Popularity** — ignores the query entirely. Deceptively strong on accuracy metrics, which is exactly why it belongs.
- **Raw cosine** — cosine similarity on the nine scaled features, no decomposition. **This is the control that matters.** The entire justification for the SVD is that projecting to a latent subspace beats using the features directly.

7.3 Results

System	HR@10	R@10	MRR	NDCG@10	Div.	Cov.
raw_cosine	0.3242	0.0351	0.0865	0.0371	0.2125	0.4698
svd_k9 [†]	0.3242	0.0351	0.0865	0.0371	0.2125	0.4698
svd_k7 [†]	0.3059	0.0322	0.0793	0.0340	0.2331	0.4875
svd_k5_whiten	0.2009	0.0224	0.0559	0.0237	0.3208	0.5032
svd_k5	0.2283	0.0241	0.0508	0.0236	0.2903	0.5052
svd_k3	0.1370	0.0153	0.0412	0.0169	0.3904	0.5128
svd_k2	0.0959	0.0086	0.0338	0.0120	0.4973	0.5189
random	0.0457	0.0057	0.0154	0.0056	1.0046	0.5202
popularity	0.0411	0.0033	0.0088	0.0036	1.0131	0.0037

Table 1: Held-out-artist evaluation, 219 queries. Higher is better throughout. [†] marks systems whose NDCG difference from **raw_cosine** is not statistically significant under a paired bootstrap test.

Coral = the no-SVD control; 'n.s.' = indistinguishable from it

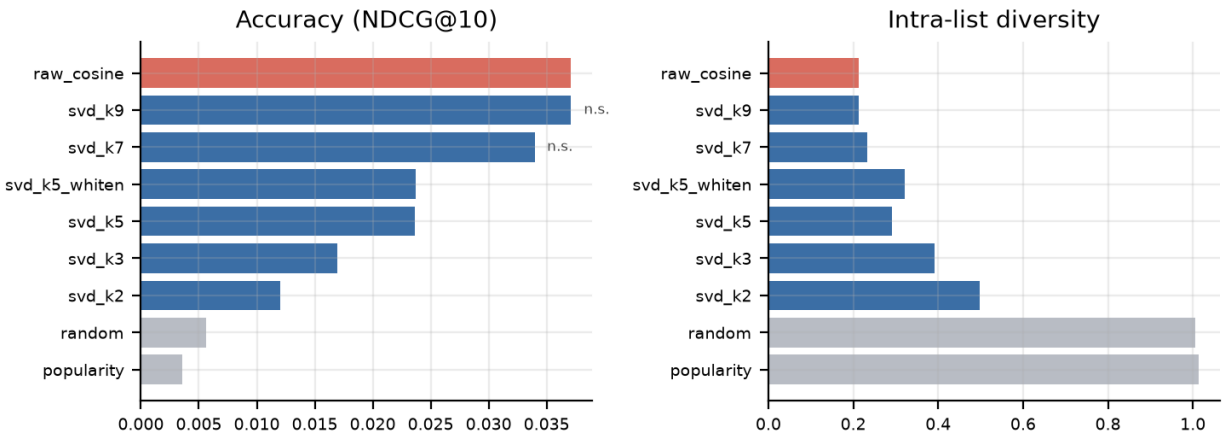


Figure 7: Accuracy and diversity. Every latent model beats random and popularity by a wide margin — the features carry real signal. None beats the no-SVD control on accuracy.

Three things follow from Table 1.

The pipeline is correct. *svd_k9* reproduces *raw_cosine* to every reported digit on every metric ($\Delta = 0$). This is the identity predicted in Section 3: at full rank the projection is a rotation, and cosine similarity is rotation-invariant. Any implementation error in scaling, decomposition or ranking would break this equality.

Truncation to $k = 7$ costs nothing measurable. *svd_k7* scores 0.0340 against 0.0371 for raw cosine. A paired bootstrap over the shared queries gives $\Delta = -0.0031$, 95% CI $[-0.0105, +0.0048]$, $p = 0.427$. The interval contains zero: on this evidence the two are indistinguishable. Meanwhile diversity rises from 0.2125 to 0.2331 and coverage from 0.4698 to 0.4875. Discarding two dimensions is free, and it broadens the results.

Aggressive truncation is significantly worse. At $k \leq 5$ every comparison against raw cosine is significant at $p < 0.01$, and accuracy falls monotonically with k . The original choice of $k = 5$ sits on the wrong side of that boundary.

The pairing matters. These systems are evaluated on identical queries, so the per-query differences share the query-difficulty variance; testing them as independent samples would badly overstate the uncertainty and could easily turn a null result into a spurious one.

7.4 Interpretation

The honest summary is that dimensionality reduction is not an accuracy technique here. It is a diversity technique. There is a regime ($k = 7$) where it is free, and below that you are trading measurable accuracy for measurable variety — which may still be the right trade for a discovery product, but it should be made deliberately rather than assumed.

This does not vindicate the original conclusion, and it does not refute the method. It replaces an untested claim with a measured trade-off.

8 Does audio-feature classification reproduce genre?

The original abstract asked whether classifying music by features would “result in similar genre-based definitions, or would the classification of music be revamped.” Recommendation quality cannot answer that. Clustering can.

We partition the latent space with k -means (k -means++ seeding, ten restarts) into as many clusters as there are genre labels, then measure agreement with metrics that are invariant to label permutation and corrected for chance: Adjusted Rand Index and Normalised Mutual Information. Purity is reported alongside but not relied upon — it can be driven to 1 simply by using more clusters.

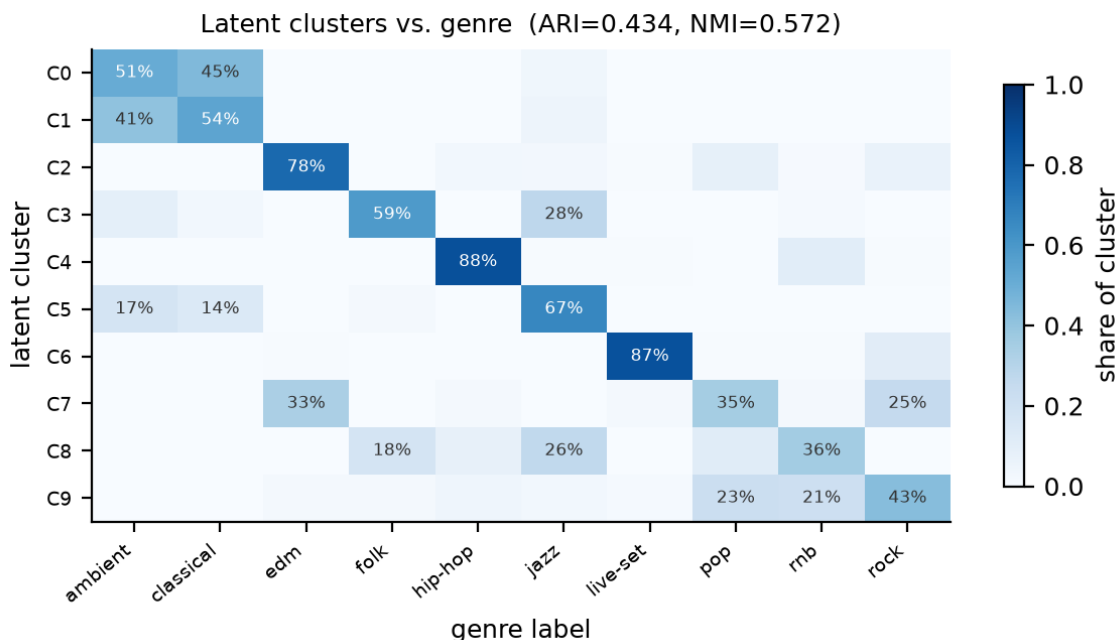


Figure 8: Each latent cluster’s composition by genre label. Rows spanning several columns are genres the audio features consider interchangeable; columns spread across rows are genres the features consider internally inconsistent.

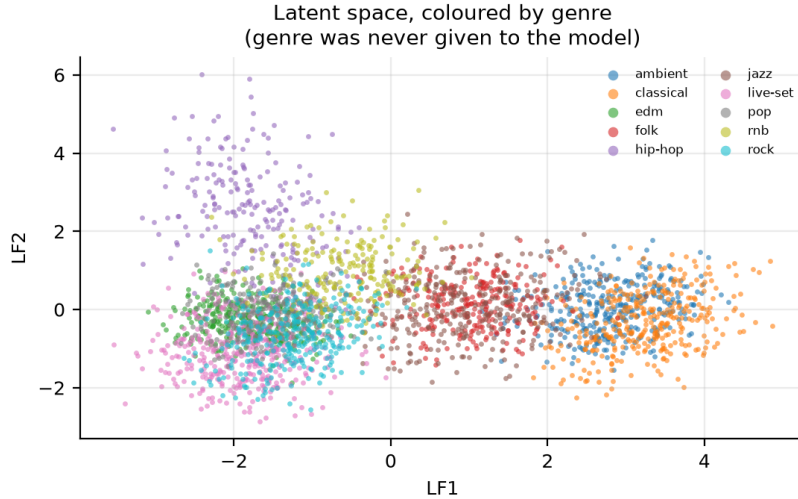


Figure 9: The catalogue in its two leading latent dimensions, coloured by a genre label the model never received.

Agreement is partial: $\text{ARI} = 0.434$, $\text{NMI} = 0.572$, $\text{purity} = 0.591$, $\text{silhouette} = 0.178$ across 10 genres. The partition partially reproduces the reference taxonomy. Neither extreme holds — the taxonomy is neither recovered nor unrelated — and the structure in Figure 8 is more informative than the scalars:

- **Cleanly separable.** Hip-hop (88% of its cluster), live recordings (87%) and EDM (78%) each occupy a cluster of their own. These are categories defined by *production*: speechiness, audience noise, synthetic timbre. Audio features measure production directly.
- **Merged.** Ambient and classical are split across two clusters, each of which is a mixture of both. The features cannot separate them at all: both are quiet, acoustic, instrumental and low-energy.
- **Dissolved.** Pop, rock and R&B do not form clusters. They are spread across three mixed groups. These are categories defined by lineage, instrumentation convention and marketing — none of which the nine features observe.

So the answer to the original question is: *revamped, but not arbitrarily*. Audio features reorganise music along an axis of production character rather than genre ancestry. Where a genre label happens to encode production, it survives; where it encodes history, it dissolves.

Caveat. This finding is measured on a synthetic catalogue whose genre profiles were specified by hand, and the merges partly reflect overlaps in those specifications — ambient and classical were given deliberately similar profiles. The methodology transfers; this particular result should be re-measured on a real corpus before being relied upon. The command to do so is one line (Section 10).

9 Defects in the original pipeline

Four defects materially affected the original results. Each was reproduced before being fixed, and each now has a regression test.

9.1 Self-recommendation

The clearest evidence is in the original paper itself. Cross-referencing its published recommendations (§4.3) against its own input playlist (§4.1):

Track	Status
<i>Slime You Out</i> (feat. SZA)	in the input playlist, returned as a recommendation
<i>All The Stars</i> (with SZA)	in the input playlist, returned
<i>I Hate U</i>	in the input playlist, returned
<i>What Lovers Do</i> (feat. SZA)	in the input playlist, returned <i>twice</i>
<i>when the party's over</i>	returned <i>twice</i>

Table 2: Duplicates and self-recommendations in the original published results. The input playlist also listed *What Lovers Do* and *All The Stars* twice each, which compounds the effect under `max` aggregation.

The cause is Section 2.1: exclusion by row index in a dataset with many rows per track. Roughly a quarter of the reported recommendations were either input tracks or repeats.

9.2 Explained variance always reported as 100%

The truncated spectrum was normalised against itself. Every configuration therefore appeared to explain all variance, making the figure uninformative and concealing how little the trailing components contributed.

9.3 Scale-dependent convergence

The eigensolver tested convergence by comparing an absolute off-diagonal sum against 10^{-10} . That sum grows with $\|A\|$, so on a 5000-row matrix the tolerance was unreachable: the solver exhausted its iteration budget every time and still exited with off-diagonal mass $\approx 8 \times 10^{-3}$. Both replacement solvers use relative tolerances.

9.4 Unrunnable entry point

The program printed `U+2713 CHECK MARK` unconditionally. On a Windows console using code page 1252 — the default on the development machine — this raises `UnicodeEncodeError` before any recommendation is printed. All terminal output now degrades to ASCII when the active encoding cannot represent a glyph, and continuous integration runs the test suite under `cp1252` so the regression cannot pass on a UTF-8 runner and still fail on a real desktop.

10 Implementation and reproducibility

Every decomposition is implemented from the underlying mathematics: Householder QR, two symmetric eigensolvers, one-sided Jacobi SVD, randomized SVD, cosine similarity, Levenshtein distance, k -means, and all ranking and agreement metrics. NumPy supplies array storage and elementwise operations; `numpy.linalg` appears only in the test suite, as ground truth.

The system is verified by 326 tests covering reconstruction accuracy, orthogonality, Eckart–Young optimality of the rank- k truncation, cross-backend agreement, and a regression test for each defect in Section 9.

Every figure, table and quoted number in this paper is produced by a single command, so the document cannot drift from the code:

```
python paper/make_figures.py
python paper/make_figures.py --data data/songs.csv % any catalogue
```

11 Limitations

- **Synthetic corpus.** All results come from a generated catalogue (see §1). The generative structure is known and influences the outcome, particularly the genre merges of Section 8.
- **Proxy ground truth.** Shared authorship is a stand-in for relevance. It penalises exactly the cross-genre discovery this system aims at.
- **Nine features.** The audio features describe production, not melody, harmony, lyrics or structure. The dissolution of pop, rock and R&B in Section 8 may say more about that limitation than about the genres.
- **No listener study.** Every claim here is offline. Whether any of it corresponds to what a person would enjoy is untested.

12 Conclusion

Adding an evaluation harness to this project produced a negative result about its central method, and that is the outcome worth keeping. Dimensionality reduction did not improve retrieval accuracy over the raw features; at $k = 7$ it was free, and below that it cost accuracy to buy diversity. The full-rank identity between the SVD recommender and plain cosine similarity makes the reason precise: the decomposition contributes nothing except through truncation, and truncation is a trade rather than a gain.

The clustering analysis answers the question the original study posed. Audio features do not reproduce the genre taxonomy, and they do not scramble it either. They reorganise music by production character — separating hip-hop, EDM and live recordings cleanly, merging ambient with classical, and dissolving pop, rock and R&B entirely. That reorganisation is the interesting object, and it is now something the repository can measure rather than assert.

References

- [1] J. Demmel and K. Veselić. Jacobi’s method is more accurate than QR. *SIAM Journal on Matrix Analysis and Applications*, 13(4):1204–1245, 1992.
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